40 mm (1.57 in) bore × 25 mm (0.984 in) rod Cylinder Force Data

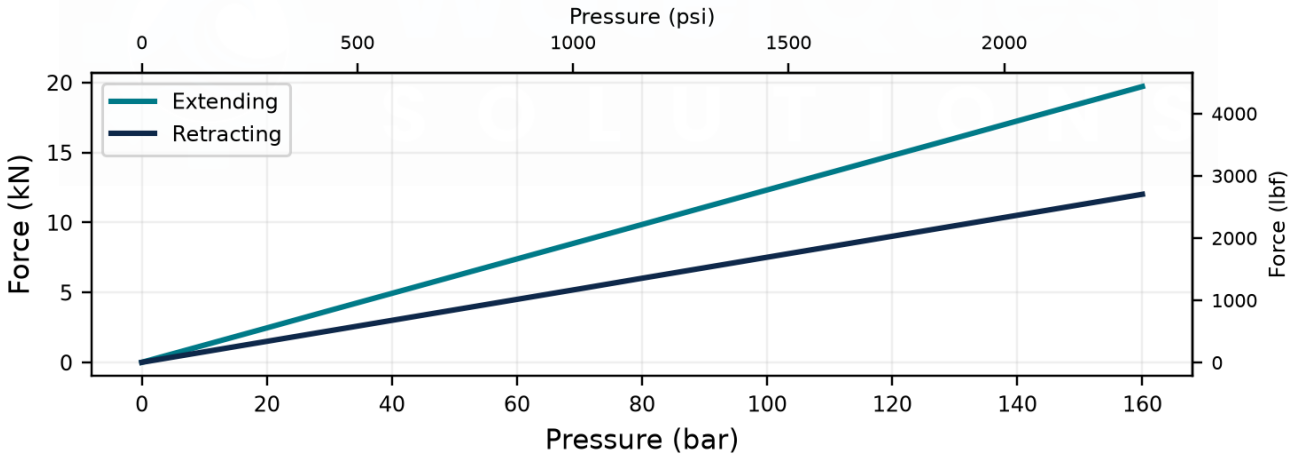
This geometry has a retraction area equal to 60.9% of its full piston area. Compare push and pull force at the same pressure, then check stroke, buckling, mounting and return-side pressure for the actual load.

Complete force output data across all operating pressures from 0 to 160 bar (0 to 2,321 psi). All values calculated with 0.98 efficiency factor.

At a glance

Parameter / value	Parameter / value
Piston Area 12.57 cm² (1.948 in²)	Ring Area 7.66 cm² (1.187 in²)
Max Extending Force 19.71 kN (4,431 lbf)	Max Retracting Force 12.01 kN (2,700 lbf)

Push and pull force



Plotted from the webpage force table, including its stated 0.98 efficiency factor.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Pressure	Force Extending (kN)	Force Retracting (kN)
0 bar (0 psi)	0.00 kN (0 lbf)	0.00 kN (0 lbf)
10 bar (145 psi)	1.23 kN (277 lbf)	0.75 kN (169 lbf)
20 bar (290 psi)	2.46 kN (553 lbf)	1.50 kN (337 lbf)
30 bar (435 psi)	3.70 kN (832 lbf)	2.25 kN (506 lbf)
40 bar (580 psi)	4.93 kN (1,108 lbf)	3.00 kN (674 lbf)
50 bar (725 psi)	6.16 kN (1,385 lbf)	3.75 kN (843 lbf)
60 bar (870 psi)	7.39 kN (1,661 lbf)	4.50 kN (1,012 lbf)
70 bar (1,015 psi)	8.62 kN (1,938 lbf)	5.25 kN (1,180 lbf)
80 bar (1,160 psi)	9.85 kN (2,214 lbf)	6.01 kN (1,351 lbf)
90 bar (1,305 psi)	11.09 kN (2,493 lbf)	6.76 kN (1,520 lbf)
100 bar (1,450 psi)	12.32 kN (2,770 lbf)	7.51 kN (1,688 lbf)
110 bar (1,595 psi)	13.55 kN (3,046 lbf)	8.26 kN (1,857 lbf)
120 bar (1,740 psi)	14.78 kN (3,323 lbf)	9.01 kN (2,026 lbf)
130 bar (1,885 psi)	16.01 kN (3,599 lbf)	9.76 kN (2,194 lbf)
140 bar (2,031 psi)	17.25 kN (3,878 lbf)	10.51 kN (2,363 lbf)
150 bar (2,176 psi)	18.48 kN (4,154 lbf)	11.26 kN (2,531 lbf)
160 bar (2,321 psi)	19.71 kN (4,431 lbf)	12.01 kN (2,700 lbf)